

Fixed-Time Leader-Following Consensus for Second-Order Multiagent Systems With Input Delay

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Abstract—This paper studies fixed-time leader-following lag consensus problem of second-order multiagent systems with input delay. Using fixed-time distributed observer, we obtain the leader's states for each followers. An extension of the Artstein's reducing transformation is employed to transform the delayed error system into a second-order system without time delay and a novel nonsingular terminal sliding mode protocol is proposed to achieve fixed-time consensus. The presented sliding mode controller can avoid singularity, eliminate chattering, and achieve exact convergence. It is mathematically proved that the presented protocol can achieve exact fixed-time leader-following lag consensus. Moreover, the upper bound of convergence time only depends on observer parameters, controller parameters, network parameters, and delay time, which makes it possible to determine the convergence time offline regardless of initial condition. The presented protocol is applied to coordinated lag tracking control of single-link robotic manipulators and the results validate the effectiveness of the proposed fixed-time protocol.

Index Terms—Fixed-time consensus, input delay, lag consensus, leader-following consensus, multiagent system, nonsingular terminal sliding mode.

I. INTRODUCTION

RECENTLY, distributed coordination of multiagent systems has attracted great attention from control community (see [1]–[20], [25]–[41], and references therein) and has found wide applications in multirobot systems tracking [6], [7], power systems [8], [9], and spacecraft formation flying [10]. This is due to the fact that a team of agents can work cooperatively to accomplish some complex tasks that cannot be achieved by a

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single agent. An important issue in distributed coordination of multiagent systems is to design distributed protocols such that all the agents reach an agreement on their common states via local interaction, which is referred to as consensus problem.

In the study of consensus problem, convergence speed is an important performance index. It has been shown that algebraic connectivity (or the second smallest eigenvalue) determines the convergence rate of consensus for multiagent system [11]. Specifically, faster convergence rate requires larger algebraic connectivity. This finding motivates some research works to improve the consensus by increasing the algebraic connectivity [12], [13]. Although by maximizing the algebraic connectivity, one can improve convergence rate via linear protocol, the consensus can never be reached within finite time. For some practical applications where control accuracy is crucial, it is required that the convergence should be reached within finite time. Moreover, compared with asymptotical consensus, finite-time consensus has better disturbance rejection property and stronger robustness against uncertainties. Therefore, finite-time consensus problem has been proposed to achieve high speed convergence. Cortés [14] proposed the normalized and signed gradient descent flow of a differentiable function to solve finite-time network consensus problem. With the help of homogeneous method, Wang and Hong [15] and Zhao *et al.* [16] presented finite-time consensus protocol for second-order multiagent system. Based on adding a power integrator method, Li *et al.* [17] constructed finite-time consensus algorithm for leaderless and leader-follower multiagent system with external disturbance. Using terminal sliding mode technique, Khoo *et al.* [18] investigated finite-time consensus problem of multirobot system. In [19], Jiang and Wang designed protocol to solve finite-time average consensus problem with respect to monotonic function. Xiao *et al.* [20] investigated finite-time consensus problem for multiagent system with switching topology. It is worth noting that the convergence time of finite-time protocol depends on initial condition. However, in many practical applications, it is always hard to obtain accurate information of initial condition, which may bring difficulties in giving an accurate estimate of settling time. In addition, the convergence time of finite-time consensus grows unboundedly with the increase of the initial condition.

Fixed-time stability [21] is an extension of finite-time stability. The convergence time of finite-time stability depends on

initial condition and grows unboundedly with the increase of the initial condition. Different from finite-time stability, fixed-time stability can guarantee convergence within finite time upper bounded by a constant independent of initial condition, which facilitates convergence time estimation and controller design to satisfy the requirement of severe settling time constraint. Due to this attractive feature, fixed-time stability has been applied to design uniform robust exact differentiator [22], design power system stable controller [23], [24] and address consensus problem for multiagent system [25]–[31]. Fixed-time consensus problem has been solved for multiagent systems in a network with undirected topology [25], directed topology [26], and directed and switching topology [27]. In [28] and [29], fixed-time robust consensus problem was addressed for multiagent system with unknown dynamics and external disturbances. In [30], fixed-time coordinate tracking problem with relative output and state measurement was studied for multiagent system with input uncertainties. In [31], fixed-time consensus was extended to multiagent system with mixed cooperative and antagonistic interactions. However, to the best of our knowledge, there is no literature reporting fixed-time consensus in the presence of time delay.

Time delay is frequently encountered in multiagent systems because of nonideal communication channels and processing after receipt. The presence of time delay may deteriorate the performance of controlled system and lead to instability. Therefore, consensus of multiagent systems with time delay has been extensively studied in many literatures (for example, see [32]–[41] and references therein). However, there is scarce literature reporting fixed-time lag consensus of multiagent systems with time delay. The lag consensus means that the states of the followers are behind the leader at different lag time, so as to avoid congestion [42], [43]. For time-delayed systems, Lyapunov–Krasovskii functionals and Lyapunov–Razumikhin functions are effective way to deal with convergence and stability problem. Unfortunately, these methods cannot be applied to study finite-time consensus problem for time-delayed systems since it is difficult to find appropriate Lyapunov function [44]. Artstein [45] proposed an invertible transformation that can transform the linear systems with delayed control input into equivalent delay-free systems and demonstrated that controllability, stabilization, and optimization problems can be investigated using equivalent systems. Therefore, this transformation facilitates many control design and has been extended to feedback control [46], optimal control [47], and finite-time control [44]. It can also be extended to fixed-time consensus protocol design.

In this paper, fixed-time leader-following lag consensus problem for second-order multiagent system with input delay is investigated. To solve the problem, we first design fixed-time distributed observer to obtain the leader's states. An extension of the Artstein's reducing transformation is used to transform the time-delayed error system into a second-order system without time delay and a novel nonsingular fixed-time terminal sliding mode protocol is constructed to achieve fixed-time consensus. Compared with the existing literatures, the main contributions of this paper can be summarized as the following

three aspects. First, fixed-time lag consensus problem for time-delayed multiagent systems has been addressed and to the best of our knowledge, no one has extended fixed-time consensus to time-delayed multiagent systems and no one has addressed fixed-time lag consensus problem. Second, a novel fixed-time nonsingular terminal sliding mode controller is constructed, which can effectively overcome singularity, eliminate chattering and guarantee exact convergence. While the fixed-time sliding mode controller presented in [30] cannot eliminate chattering and convergence error simultaneously. Third, the upper bound of convergence time can be estimated in advance, which is independent of initial condition but only depends on observer parameters, controller parameters, network parameters, and delay time.

The rest of this paper is organized as follows. Section II formulates the problem and reviews preliminary knowledge necessary throughout the paper. Main results of this paper are presented in Section III. The proposed scheme is further discussed in Section IV. Simulation results verifying the effectiveness of proposed controller are given in Section V. Finally, the conclusion is drawn in Section VI.

II. PROBLEM FORMULATION AND PRELIMINARIES

A. Graph Theory

Let $G(A) = \{V, E, A\}$ be a graph with a node set $V(G) = \{v_0, v_1, \dots, v_N\}$ with index 0 being leader node, an edge set $E \subseteq V \times V$ and an adjacent matrix $A = [a_{ij}] \in R^{N \times N}$. For directed graph, an edge (v_i, v_j) denotes directed communication link from node v_i to node v_j , but not necessarily vice versa. In contrast, for undirected graph, $(v_i, v_j) \in E$ implies $(v_j, v_i) \in E$, i.e., the nodes v_i and v_j can send information to each other. If $(v_i, v_j) \in E$, node v_i is called the neighbor node of node v_j or v_i and v_j are said to be adjacent. The index set $N_j = \{i : (v_i, v_j) \in E\}$ includes the nodes index of all neighbors of v_j . For undirected graph, $A = [a_{ij}]$ is the adjacency matrix with nonnegative elements $a_{ij} = a_{ji} \geq 0$. Moreover, $a_{ij} > 0$ if $(v_i, v_j) \in E$, $a_{ij} = 0$ if $(v_j, v_i) \notin E$, and $a_{ii} = 0$ for $i \in I_N = 1, \dots, N$. Since the follower cannot influence the leader, we have $a_{0i} = 0, i = 1, \dots, N$, $a_{i0} > 0$ if the follower has direct link with the leader and $a_{i0} = 0$ otherwise. For directed graph, the definition of the adjacency matrix $A = [a_{ij}]$ is similar except $a_{ij} = a_{ji}$. The Laplacian matrix of $G(A)$ is defined as $L = [l_{ij}]$, where $l_{ii} = \sum_{j \neq i} a_{ij}$ and $l_{ij} = -a_{ij}$ where $i \neq j$. An induced subgraph G_s of G is a graph such that $V(G_s) \subset V(G)$ and $E(G_s) \subset E(G)$. Let L_s denote the Laplacian matrix of subgraph G_s , $B = \text{diag}\{a_{10}, \dots, a_{N0}\}$ and define $H = L_s + B$. H has an important property that is given in Lemma 1. A sequence of distinct vertices $\{v_i, \dots, v_j\}$ is called a path from node v_i to node v_j if the consecutive vertices of the sequence are adjacent. A directed graph is called strongly connected if and only if for any nodes v_i, v_j , there exists a directed path from i to j . An undirected graph is called connected if there exists a path for any two nodes.

Lemma 1 (see [48]): If the subgraph G_s is connected and undirected and each follower has a path to the leader in the graph G , then H is symmetric and positive definite.

B. Fixed-Time Stability

Consider the following differential equation system:

$$\dot{x}(t) = f(x(t)), x(0) = x_0 \quad (1)$$

where $x \in R$ and $f : R_+ \times R^n \rightarrow R^n$ is a nonlinear function. Suppose that the origin is an equilibrium point of (1).

Definition 1 (see [28], [49]): The origin of system (1) is a finite-time stable equilibrium if the origin is Lyapunov stable and there exists a function $T : R^n \rightarrow R^+$, called the settling time function, such that for every $x_0 \in R^n$, the solution $x(t, x_0)$ of system (1) is defined on $[0, T(x_0))$, $x(t, x_0) \in R^n$, for all $t \in [0, T(x_0))$, and $\lim_{t \rightarrow T(x_0)} x(t, x_0) = 0$.

Definition 2 (see [21]): The origin of system (1) is said to be a fixed-time stable equilibrium point if it is globally finite-time stable with bounded convergence time $T(x_0)$, that is, there exists a bounded positive constant $T_{\max}$ such that $T(x_0) < T_{\max}$ satisfies.

C. Problem Formulation

Consider a group of $N + 1$ agents, where the agent whose index is 0 is assigned as the “leader” and the other agents indexed by $i = 1, \dots, N$ are called “follower.” The dynamics of each “follower” can be described as

$$\begin{cases} \dot{x}_{1i} = x_{2i} \\ \dot{x}_{2i} = u_{2i}(t - \tau_{\text{coni}}) \end{cases} \quad (2)$$

where $x_{1i}, x_{2i}, u_{2i}, \tau_{\text{coni}} (i = 1, \dots, N)$ denote the position-like state, velocity-like state, control input, and constant input delay of follower agent i .

The leader’s dynamics can be expressed as

$$\begin{cases} \dot{x}_{10} = x_{20} \\ \dot{x}_{20} = u_{20}(t) \end{cases} \quad (3)$$

where x_{10}, x_{20}, u_{20} represent the position-like state, velocity-like state, and control input of leader agent 0.

Assumption 1: The control input of the leader agent and its derivative are bounded, that is, there exist positive constants ρ_1 and ρ_2 such that $|u_{20}| \leq \rho_1, |\dot{u}_{20}| \leq \rho_2$.

Assumption 2: The input $u_{20}(t)$ of the leader is unavailable to all the followers but its upper bounds ρ_1 and ρ_2 are available to all the agents.

Assumption 3: The subgraph G_s defined on the N followers is a connected undirected graph and each follower has a path to the leader in the graph G .

Remark 1: Assumption 2 is reasonable because it is more costly and difficult for followers to obtain the information of the leader’s input than the leader’s states but the upper bound of the leader’s input can be obtained through a prior knowledge of the physical systems [50]. Additionally, Assumption 3 is important because without this assumption, leader-following consensus cannot be achieved [51].

This multiagent system is said to solve fixed-time leader-following lag consensus problem if there exists a settling time $T(x(0)) \in [0, \infty)$ whose upper bound is independent of initial state $x(0) \in R^N$, i.e., $\exists T_{\max} > 0 : T(x(0)) < T_{\max}, \forall x(0) \in R^N$, such that $\lim_{t \rightarrow T(x(0))} |x_{ij}(t) - x_{i0}(t - \tau_{\text{conj}})| = 0$ for $i =$

$1, 2, j = 1, \dots, N$ and $x_{ij}(t) = x_{i0}(t - \tau_{\text{conj}})$ for $t \geq T(x(0))$ and $\tau_{\text{conj}} > 0$. As shown in [42] and [43], lag consensus can avoid congestion. This paper will design fixed-time protocol to achieve fixed-time leader-following lag consensus for multiagent system (2) and (3).

D. Mathematical Lemmas

In order to achieve fixed-time leader-following lag consensus for multiagent system, we first introduce some useful lemmas that are necessary for protocol design.

Lemma 2 (see [29]): Consider the following system:

$$\dot{y} = -\alpha y^{p/q} - \beta y^{m/n}, y(0) = y_0 \quad (4)$$

where m, n, p, q are positive odd integers satisfying $m > n, p < q, \alpha, \beta$ are positive real numbers. Then, the equilibrium point of system (4) is globally finite-time stable with the settling time upper bounded by

$$T < \frac{1}{\alpha} \frac{q}{q-p} + \frac{1}{\beta} \frac{n}{m-n}. \quad (5)$$

Corollary 1: The trajectory of system (4) starting from any initial state will converge to a compact set $W = \{y : y < r, 0 < r < 1\}$ within finite time upper bounded by

$$T < \frac{1 - r^{1-p/q}}{\alpha} \frac{q}{q-p} + \frac{1}{\beta} \frac{n}{m-n}. \quad (6)$$

Proof: Define a new variable as $w = y^{1-p/q}$ and the expression of system (4) can be rewritten as

$$\dot{w} = -\frac{q-p}{q} (\alpha + \beta w^{\frac{m/n-p/q}{1-p/q}}). \quad (7)$$

Let $\gamma = (m-n)q/((q-p)n)$ and the expression (7) becomes

$$\dot{w} = -\frac{q-p}{q} (\alpha + \beta w^{1+\gamma}). \quad (8)$$

Solve the equation (8) for convergence time and the system will converge to a compact set $W = \{y : y < r, 0 < r < 1\}$ within finite time upper bounded by

$$\begin{aligned} \lim_{w(0) \rightarrow \infty} T(w(0)) &= \frac{q}{q-p} \int_{r^{1-p/q}}^{+\infty} \frac{1}{\alpha + \beta w^{1+\gamma}} dw \\ &< \frac{q}{q-p} \left(\int_1^{+\infty} \frac{1}{\beta w^{1+\gamma}} dw + \int_{r^{1-p/q}}^1 \frac{1}{\alpha} dw \right) \\ &= \frac{1 - r^{1-p/q}}{\alpha} \frac{q}{q-p} + \frac{1}{\beta} \frac{n}{m-n}. \end{aligned} \quad (9)$$

The proof is completed.

Lemma 3 (see [21]): If there exists a continuous radially unbounded function $V : R^N \rightarrow R_+ \cup 0$ satisfying (i) $V(x) = 0 \Rightarrow x = 0$; (ii) $\dot{V}(x(t)) \leq -(\alpha V^p(x(t)) + \beta V^q(x(t)))^k$ for some $\alpha, \beta, p, q, k > 0, pk < 1, qk > 1$ and for any solution $x(t)$, then the origin of system (1) is globally finite-time stable and the convergence time is upper bounded by

$$T \leq \frac{1}{\alpha^k(1-pk)} + \frac{1}{\beta^k(qk-1)}. \quad (10)$$

Lemma 4: For variable x with initial condition x_0 , if the dynamics of x satisfies

$$\dot{x} = -k_1 x - k_2 \text{sign}(x), \quad (11)$$

where k_1 and k_2 are positive real numbers. Then, the origin of system (11) is finite-time stable with the settling time satisfying

$$T = \frac{1}{k_1} \ln \left(1 + \frac{k_1 x_0}{k_2} \right). \quad (12)$$

Proof: Consider the following Lyapunov function candidate:

$$V = \frac{1}{2} x^2. \quad (13)$$

The derivative of V can be derived as

$$\dot{V} = -k_1 x^2 - k_2 |x| = -2k_1 V - k_2 (2V)^{1/2}. \quad (14)$$

Let $\eta = V^{1/2}$ and (14) can be rewritten as

$$\dot{\eta} = -k_1 \eta - k_2 / \sqrt{2}. \quad (15)$$

Equation (15) can be rewritten as $-d\eta / (k_1 \eta + k_2 / \sqrt{2}) = dt$. Solve (15) and the settling time can be estimated as

$$T = \int_0^T dt = - \int_{x_0/\sqrt{2}}^0 \frac{1}{k_1 \eta + k_2 / \sqrt{2}} d\eta = \frac{1}{k_1} \ln \left(1 + \frac{k_1 x_0}{k_2} \right). \quad (16)$$

The proof is completed.

Lemma 5 (see [52]): For any nonnegative real numbers $\xi_1, \xi_2, \dots, \xi_N$ and $0 < p \leq 1$, the following inequality holds:

$$\sum_{i=1}^N \xi_i^p \geq \left(\sum_{i=1}^N \xi_i \right)^p. \quad (17)$$

Lemma 6 (see [52]): For any nonnegative real numbers $\xi_1, \xi_2, \dots, \xi_N$ and $p > 1$, the following inequality holds:

$$\sum_{i=1}^N \xi_i^p \geq N^{1-p} \left(\sum_{i=1}^N \xi_i \right)^p. \quad (18)$$

III. MAIN RESULTS

Since the leader's states x_{10} and x_{20} are only available to the followers that have direct communication link to the leader, it is needed to estimate the leader's states and fixed-time distributed observer is constructed as follows:

$$\begin{cases} \dot{\xi}_i = \eta_i - \alpha \text{sig} \left(\sum_{j=0}^N a_{ij} (\xi_i - \xi_j) \right)^{m/n} \\ \quad - \beta \text{sig} \left(\sum_{j=0}^N a_{ij} (\xi_i - \xi_j) \right)^{p/q} \\ \dot{\eta}_i = -\alpha \text{sig} \left(\sum_{j=0}^N a_{ij} (\eta_i - \eta_j) \right)^{m/n} \\ \quad - \beta \text{sig} \left(\sum_{j=0}^N a_{ij} (\eta_i - \eta_j) \right)^{p/q} \\ \quad - \lambda \text{sign} \left(\sum_{j=0}^N a_{ij} (\eta_i - \eta_j) \right) \end{cases} \quad (19)$$

where ξ_i and η_i ($i = 1, \dots, N$) are state variables of the observer for the i th agent, $\xi_0 = \eta_0 = 0$, α, β, λ are observer gains that satisfy $\lambda \geq \rho_1$, m, n, p, q are positive constant satisfying $m > n$ and $p < q$, $\text{sig}(\cdot)^\alpha = |\cdot|^\alpha \text{sign}(\cdot)$.

Remark 2: In this paper, we study lag consensus problem and need the leader's states at $t - \tau_{\text{con}i}$ rather than the tracking errors at t . This is why we design the observer (19) instead of the observer presented in [30].

Theorem 1: Suppose that the leader's control input u_{20} satisfies Assumption 1 and Assumptions 2 and 3 hold. Distributed observer (19) can estimate the leader's states x_{10} and x_{20} within globally bounded time, that is, we have $\xi_i = x_{10}$ and $\eta_i = x_{20}$ after $T < 2 \left(\frac{1}{\alpha N^{\frac{1-m/n}{2}} \lambda_{\min}(H)} \frac{n}{m-n} + \frac{1}{\beta \lambda_{\min}(H)} \frac{q}{q-p} \right)$.

Proof: Define $\tilde{\xi}_i = \xi_i - x_{10}$, $\tilde{\eta}_i = \eta_i - x_{20}$, and we have

$$\begin{cases} \dot{\tilde{\xi}}_i = \tilde{\eta}_i - \alpha \text{sig} \left(\sum_{j=0}^N a_{ij} (\tilde{\xi}_i - \tilde{\xi}_j) \right)^{m/n} \\ \quad - \beta \text{sig} \left(\sum_{j=0}^N a_{ij} (\tilde{\xi}_i - \tilde{\xi}_j) \right)^{p/q} \\ \dot{\tilde{\eta}}_i = -\alpha \text{sig} \left(\sum_{j=0}^N a_{ij} (\tilde{\eta}_i - \tilde{\eta}_j) \right)^{m/n} \\ \quad - \lambda \text{sign} \left(\sum_{j=0}^N a_{ij} (\tilde{\eta}_i - \tilde{\eta}_j) \right) \\ \quad - \beta \text{sig} \left(\sum_{j=0}^N a_{ij} (\tilde{\eta}_i - \tilde{\eta}_j) \right)^{p/q} - u_{20}(t) \end{cases}. \quad (20)$$

Let $\tilde{\eta} = [\tilde{\eta}_1, \dots, \tilde{\eta}_N]^T$ and consider the Lyapunov function $V = \frac{1}{2} \tilde{\eta}^T H \tilde{\eta}$. Since the communication network topology satisfies Assumption 3 and according to Lemma 1, V is a positive definite and radially unbounded function of $\tilde{\eta}$. The symmetric of H gives that

$$\frac{dV}{d\tilde{\eta}_i} = \sum_{j=0}^N a_{ij} (\tilde{\eta}_i - \tilde{\eta}_j). \quad (21)$$

Thus, using Lemmas 5 and 6, we have that the derivative of V satisfies

$$\begin{aligned} \dot{V} &= \sum_{i=1}^N \frac{dV}{d\tilde{\eta}_i} \dot{\tilde{\eta}}_i \\ &\leq -\alpha \sum_{i=1}^N \left| \sum_{j=0}^N a_{ij} (\tilde{\eta}_i - \tilde{\eta}_j) \right|^{m/n+1} \\ &\quad - \beta \sum_{i=1}^N \left| \sum_{j=0}^N a_{ij} (\tilde{\eta}_i - \tilde{\eta}_j) \right|^{p/q+1} \\ &\quad - (\lambda - \rho_1) \left| \sum_{j=0}^N a_{ij} (\tilde{\eta}_i - \tilde{\eta}_j) \right| \\ &\leq -\alpha \sum_{i=1}^N \left(\left| \sum_{j=0}^N a_{ij} (\tilde{\eta}_i - \tilde{\eta}_j) \right|^2 \right)^{\frac{m/n+1}{2}} \\ &\quad - \beta \sum_{i=1}^N \left(\left| \sum_{j=0}^N a_{ij} (\tilde{\eta}_i - \tilde{\eta}_j) \right|^2 \right)^{\frac{p/q+1}{2}} \\ &\leq -\alpha N^{\frac{1-m/n}{2}} \left(\sum_{i=1}^N \left(\sum_{j=0}^N a_{ij} (\tilde{\eta}_i - \tilde{\eta}_j) \right)^2 \right)^{\frac{m/n+1}{2}} \\ &\quad - \beta \left(\sum_{i=1}^N \left(\sum_{j=0}^N a_{ij} (\tilde{\eta}_i - \tilde{\eta}_j) \right)^2 \right)^{\frac{p/q+1}{2}}. \end{aligned} \quad (22)$$

Since H is a positive definite matrix, there exists an invertible matrix M such that $H = M^T M$. After some simple manipulations, we have

$$\frac{\sum_{i=1}^N (\sum_{j=0}^N a_{ij} (\tilde{\eta}_i - \tilde{\eta}_j))^2}{V} = \frac{\tilde{\eta}^T H^T H \tilde{\eta}}{\frac{1}{2} \tilde{\eta}^T H \tilde{\eta}} \geq 2\lambda_{\min}(H). \quad (23)$$

Substituting (23) into the last inequality of (22), we have

$$\dot{V} \leq -\alpha N^{\frac{1-m/n}{2}} (2\lambda_{\min}(H)V)^{\frac{m/n+1}{2}} - \beta (2\lambda_{\min}(H)V)^{\frac{p/q+1}{2}}. \quad (24)$$

Let $\chi_1 = \sqrt{2\lambda_{\min}(H)V}$ and (24) becomes

$$\dot{\chi}_1 \leq -\alpha N^{\frac{1-m/n}{2}} \lambda_{\min}(H) \chi_1^{m/n} - \beta \lambda_{\min}(H) \chi_1^{p/q}. \quad (25)$$

According to Lemma 2, we can see that distributed observer (19) can estimate the leader's states x_{20} within finite time and the settling time T_1 satisfies

$$T_1 < \frac{1}{\alpha N^{\frac{1-m/n}{2}} \lambda_{\min}(H)} \frac{n}{m-n} + \frac{1}{\beta \lambda_{\min}(H)} \frac{q}{q-p}. \quad (26)$$

After the convergence of $\tilde{\eta}_i$, the dynamics of $\tilde{\xi}_i$ becomes $\dot{\tilde{\xi}}_i = -\alpha \text{sig}(\sum_{j=0}^N a_{ij} (\tilde{\xi}_i - \tilde{\xi}_j))^{m/n} - \beta \text{sig}(\sum_{j=0}^N a_{ij} (\tilde{\xi}_i - \tilde{\xi}_j))^{p/q}$. Select the candidate Lyapunov function $V' = \frac{1}{2} \tilde{\xi}^T H \tilde{\xi}$ where $\tilde{\xi} = [\tilde{\xi}_1, \dots, \tilde{\xi}_N]^T$, follow the same process as above, and we have that the convergence time for $\tilde{\xi}_i$ is also T_1 . Therefore, it can be concluded that distributed observer (19) can estimate the leader's states x_{10} and x_{20} within finite time upper bounded by

$$T = 2T_1 < 2 \left(\frac{1}{\alpha N^{\frac{1-m/n}{2}} \lambda_{\min}(H)} \frac{n}{m-n} + \frac{1}{\beta \lambda_{\min}(H)} \frac{q}{q-p} \right). \quad (27)$$

The proof is completed.

Define the tracking errors $e_{x_{1i}} = x_{1i} - x_{10}(t - \tau_{\text{coni}})$ and $e_{x_{2i}} = x_{2i} - x_{20}(t - \tau_{\text{coni}})$ and their dynamics are

$$\begin{cases} \dot{e}_{x_{1i}} = e_{x_{2i}} \\ \dot{e}_{x_{2i}} = u_{2i}(t - \tau_{\text{coni}}) - u_{20}(t - \tau_{\text{coni}}) \end{cases} \quad (28)$$

In order to transform the delayed second-order integrator (28) into a delay-free second-order integrator, an extension of Artstein's transformation to leader-following lag consensus is introduced

$$\begin{cases} e_{y_{1i}} = e_{x_{1i}} + \int_{-\tau_{\text{coni}}}^0 (-\tau_{\text{coni}} - s)(u_{2i}(t+s) - u_{20}(t+s))ds \\ e_{y_{2i}} = e_{x_{2i}} + \int_{-\tau_{\text{coni}}}^0 (u_{2i}(t+s) - u_{20}(t+s))ds. \end{cases} \quad (29)$$

After transformation, (28) becomes

$$\begin{cases} \dot{e}_{y_{1i}} = e_{y_{2i}} - \tau_{\text{coni}}(u_{2i}(t) - u_{20}(t)) \\ \dot{e}_{y_{2i}} = u_{2i}(t) - u_{20}(t) \end{cases} \quad (30)$$

Define new variables:

$$\begin{cases} e_{z_{1i}} = e_{y_{1i}} + \tau_{\text{coni}} e_{y_{2i}} \\ e_{z_{2i}} = e_{y_{2i}} \end{cases} \quad (31)$$

and we have

$$\begin{cases} \dot{e}_{z_{1i}} = e_{z_{2i}} \\ \dot{e}_{z_{2i}} = u_{2i}(t) - u_{20}(t) \end{cases} \quad (32)$$

For each follower, terminal sliding surface can be constructed as

$$s_i = e_{z_{2i}} + \alpha_1 \mu_r(e_{z_{1i}}) + \beta_1 \text{sig}(e_{z_{1i}})^{m_1/n_1} \quad (33)$$

where

$$\mu_r(e_{z_{1i}}) = \begin{cases} \text{sig}(e_{z_{1i}})^{p_1/q_1}, & \text{if } |e_{z_{1i}}| > r \\ -\beta_1/\alpha_1 \text{sig}(e_{z_{1i}})^{m_1/n_1} + k_1 e_{z_{1i}} + k_2 \text{sign}(e_{z_{1i}}) & \text{if } |e_{z_{1i}}| \leq r \end{cases} \quad (34)$$

where k_1 and k_2 are positive real numbers, $0 < r < 1$ and $k_1 \alpha_1 r + k_2 \alpha_1 = \alpha_1 r^{p_1/q_1} + \beta_1 r^{m_1/n_1}$, $\text{sig}(\cdot)^\alpha = |\cdot|^\alpha \text{sign}(\cdot)$.

To achieve fixed-time leader-following lag consensus, the controller can be designed as

$$u_{2i}(t) = \begin{cases} -\alpha_1 p_1/q_1 |e_{z_{1i}}|^{p_1/q_1-1} e_{z_{2i}} - \beta_1 m_1/n_1 |e_{z_{1i}}|^{m_1/n_1-1} e_{z_{2i}} \\ -(\alpha_1 |s_i|^{m_2/n_2} + \beta_1 |s_i|^{m_3/n_3})^{n_1/m_1} \text{sign}(s_i) + u_{20}(t) & \text{if } |e_{z_{1i}}| > r \\ -(\alpha_1 |s_i|^{m_2/n_2} + \beta_1 |s_i|^{m_3/n_3})^{n_1/m_1} \text{sign}(s_i) - \alpha_1 k_1 e_{z_{2i}} \\ + u_{20}(t), & \text{if } |e_{z_{1i}}| \leq r \end{cases} \quad (35)$$

where $m_1, n_1, m_2, n_2, m_3, n_3$ are positive odd integers satisfying $1 < m_2/n_2 < m_1/n_1 < m_3/n_3$. If $e_{z_{1i}} \rightarrow 0$ and $e_{z_{2i}} \neq 0$, the singularity term $\alpha_1 p_1/q_1 |e_{z_{1i}}|^{p_1/q_1-1} e_{z_{2i}}$ in (35) will become infinite and singularity problem occurs. It is worth noting that the proposed control protocol (35) can effectively avoid singularity problem. When $|e_{z_{1i}}| > r$, we have $\alpha_1 p_1/q_1 |e_{z_{1i}}|^{p_1/q_1-1} e_{z_{2i}} \leq \alpha_1 p_1/q_1 r^{p_1/q_1-1} e_{z_{2i} \max}$, where $|e_{z_{2i}}| \leq e_{z_{2i} \max}$. This means that the control protocol (35) is bounded and singularity problem can be avoided.

Remark 3: The transformation (29) can be achieved without the information of the leader's input u_{20} . Since $\dot{x}_{20} = u_{20}$, we have $\int_{-\tau_{\text{coni}}}^0 u_{20}(t+s)ds = x_{20}(t) - x_{20}(t - \tau_{\text{coni}})$, $\int_{-\tau_{\text{coni}}}^0 (-\tau_{\text{coni}} - s)u_{20}(t+s)ds = \int_{t-\tau_{\text{coni}}}^t (-\tau_{\text{coni}} - (T-t))u_{20}(T)dT = (t - \tau_{\text{coni}}) \int_{t-\tau_{\text{coni}}}^t u_{20}(T)dT - \int_{t-\tau_{\text{coni}}}^t T u_{20}(T)dT = (t - \tau_{\text{coni}})(x_{20}(t) - x_{20}(t - \tau_{\text{coni}})) - (t x_{20}(t) - (t - \tau_{\text{coni}})x_{20}(t - \tau_{\text{coni}})) + \int_{t-\tau_{\text{coni}}}^t x_{20}(T)dT$. Therefore, only the history values of the control input of the i -th follower, the obtained leader's states and their corresponding time on time interval $[t - \tau_{\text{coni}}, t]$ are required to be stored to compute transformation (29). In addition, robust differentiator, such as uniform robust exact differentiator presented in [22], can be used to give exact derivative of x_{20} , i.e., u_{20} . This differentiator shows robustness property against perturbations and noise. According to [22], the differentiation accuracy is determined by the sampling time interval and sufficiently small sampling time interval means sufficiently high accuracy.

Theorem 2: For error system (32), the terminal sliding mode protocol (35) can achieve fixed-time leader-following lag consensus within finite time upper bounded by

$$T < \frac{1}{\alpha_1^{n_1/m_1}(1-m_2n_1/n_2m_1)} + \frac{1}{\beta_1^{n_1/m_1}(m_3n_1/n_3m_1-1)} + \frac{1-r^{1-p_1/q_1}}{\alpha_1} \frac{q_1}{q_1-p_1} + \frac{1}{\beta_1} \frac{n_1}{m_1-n_1} + \frac{1}{\alpha_1 k_1} \ln \left(1 + \frac{k_1 r}{k_2} \right) + \tau_{\text{coni}} \quad (36)$$

where $m_1, n_1, m_2, n_2, m_3, n_3$ are positive odd integers satisfying $1 < m_2/n_2 < m_1/n_1 < m_3/n_3$.

Proof: Differentiating sliding surface (33) with respect to time, one has

$$\dot{s}_i = \begin{cases} u_{2i}(t) - u_{20}(t) + \alpha_1 p_1 / q_1 |e_{z_{1i}}|^{p_1/q_1-1} e_{z_{2i}} \\ + \beta_1 m_1 / n_1 |e_{z_{1i}}|^{m_1/n_1-1} e_{z_{2i}}, & \text{if } |e_{z_{1i}}| > r \\ u_{2i}(t) - u_{20}(t) + \alpha_1 k_1 e_{z_{2i}} + 2\alpha_1 k_2 \delta(e_{z_{1i}}) e_{z_{2i}} \\ & \text{if } |e_{z_{1i}}| \leq r \end{cases} \quad (37)$$

where $\delta(x)$ is Dirac function.

Select the following Lyapunov function candidate

$$V_1 = |s_i|. \quad (38)$$

If $e_{z_{1i}} \neq 0$, the time derivative of Lyapunov function V_1 can be obtained as follows:

$$\begin{aligned} \dot{V}_1 &= \dot{s}_i \text{sign}(s_i) \\ &\leq -(\alpha_1 |s_i|^{m_2/n_2} + \beta_1 |s_i|^{m_3/n_3})^{n_1/m_1} \\ &\leq -(\alpha_1 V_1^{m_2/n_2} + \beta_1 V_1^{m_3/n_3})^{n_1/m_1}. \end{aligned} \quad (39)$$

Let us consider the case when $e_{z_{1i}} = 0$ and $e_{z_{2i}} \neq 0$. This case may occur before the system achieves consensus. Since $\dot{e}_{z_{1i}} = e_{z_{2i}}$, $e_{z_{1i}}$ will leave the origin immediately. Therefore, before the error system states reach sliding surface, the system lies in the origin of $e_{z_{1i}}$ transiently and the time that the system lies in the origin of $e_{z_{1i}}$ is negligible. After that time, the error system continues moving toward the sliding surface. According to Lemma 3, the error system will reach the sliding surface within finite time upper bounded by

$$T_1 \leq \frac{1}{\alpha_1^{n_1/m_1}(1-m_2n_1/n_2m_1)} + \frac{1}{\beta_1^{n_1/m_1}(m_3n_1/n_3m_1-1)}. \quad (40)$$

After T_1 , error system states are on sliding surface $s_i = 0$ and the dynamics of the error system becomes

$$\begin{aligned} \dot{e}_{z_{1i}} &= e_{z_{2i}} \\ &= \begin{cases} -\alpha_1 \text{sig}(e_{z_{1i}})^{p_1/q_1} - \beta_1 \text{sig}(e_{z_{1i}})^{m_1/n_1}, & \text{if } |e_{z_{1i}}| > r \\ -\alpha_1 k_1 e_{z_{1i}} - \alpha_1 k_2 \text{sign}(e_{z_{1i}}), & \text{if } |e_{z_{1i}}| \leq r. \end{cases} \end{aligned} \quad (41)$$

When $|e_{z_{1i}}| > r$, the Lyapunov function candidate is selected as

$$V_2 = |e_{z_{1i}}|. \quad (42)$$

Differentiating Lyapunov function V_2 with respect to time yields

$$\begin{aligned} \dot{V}_2 &= \dot{e}_{z_{1i}} \text{sign}(e_{z_{1i}}) = -\alpha_1 |e_{z_{1i}}|^{p_1/q_1} - \beta_1 |e_{z_{1i}}|^{m_1/n_1} \\ &= -\alpha_1 V_2^{p_1/q_1} - \beta_1 V_2^{m_1/n_1}. \end{aligned} \quad (43)$$

According to Corollary 1, system will converge to a compact set $W = \{e_{z_{1i}} : |e_{z_{1i}}| < r, 0 < r < 1\}$ within finite time upper bounded by

$$T_2 < \frac{1-r^{1-p_1/q_1}}{\alpha_1} \frac{q_1}{q_1-p_1} + \frac{1}{\beta_1} \frac{n_1}{m_1-n_1}. \quad (44)$$

When $|e_{z_{1i}}| \leq r$, the Lyapunov function candidate is constructed as

$$V_3 = \frac{1}{2} e_{z_{1i}}^2. \quad (45)$$

The time derivative of Lyapunov function V_3 can be derived as

$$\dot{V}_3 = -\alpha_1 k_1 e_{z_{1i}}^2 - \alpha_1 k_2 |e_{z_{1i}}| = -2\alpha_1 k_1 V_3 - \alpha_1 k_2 (2V_3)^{1/2}. \quad (46)$$

According to Lemma 4, error variable $e_{z_{1i}}$ will reach the origin within finite time

$$T_3 = \frac{1}{\alpha_1 k_1} \ln \left(1 + \frac{k_1 r}{k_2} \right). \quad (47)$$

When $e_{z_{1i}}$ settles down to the origin along the sliding surface $s_i = 0$, $e_{z_{2i}}$ also converges to zero. When $e_{z_{1i}} = 0$ and $e_{z_{2i}} = 0$, according to (35), $u_{2i} = u_{20}$. Then, $\int_{-\tau_{\text{coni}}}^0 (-\tau_{\text{coni}} - s)(u_{2i}(t+s) - u_{20}(t+s))ds$ and $\int_{-\tau_{\text{coni}}}^0 (u_{2i}(t+s) - u_{20}(t+s))ds$ become zero after $T_1 + T_2 + T_3 + \tau_{\text{coni}}$. Therefore, exact fixed-time leader-following lag consensus can be achieved within finite time upper bounded by

$$T < \frac{1}{\alpha_1^{n_1/m_1}(1-m_2n_1/n_2m_1)} + \frac{1}{\beta_1^{n_1/m_1}(m_3n_1/n_3m_1-1)} + \frac{1-r^{1-p_1/q_1}}{\alpha_1} \frac{q_1}{q_1-p_1} + \frac{1}{\beta_1} \frac{n_1}{m_1-n_1} + \frac{1}{\alpha_1 k_1} \ln \left(1 + \frac{k_1 r}{k_2} \right) + \tau_{\text{coni}}. \quad (48)$$

The proof is completed.

From Theorems 1 and 2, we have the following result:

Theorem 3: Consider a multiagent system consisting of one leader whose dynamics is described by (3) with the control input satisfying Assumption 1 and a group of followers whose dynamics are described by (2). Suppose that Assumptions 2 and 3 hold. Then, the distributed control scheme consisting of distributed observer (19) and terminal sliding mode protocol (35) can achieve fixed-time leader-following lag consensus for

the multiagent system within finite time upper bounded by

$$T < 2 \left(\frac{1}{\alpha N^{\frac{1-m}{2}} \lambda_{\min}(H)} \frac{n}{m-n} + \frac{1}{\beta \lambda_{\min}(H)} \frac{q}{q-p} \right) + \frac{1}{\alpha_1^{n_1/m_1} (1-m_2 n_1/n_2 m_1)} + \frac{1}{\beta_1^{n_1/m_1} (m_3 n_1/n_3 m_1 - 1)} + \frac{1-r^{1-p_1/q_1}}{\alpha_1} \frac{q_1}{q_1-p_1} + \frac{1}{\beta_1} \frac{n_1}{m_1-n_1} + \frac{1}{\alpha_1 k_1} \ln \left(1 + \frac{k_1 r}{k_2} \right) + \max(\tau_{\text{coni}}). \quad (49)$$

If the control input of the leader agent is zero, i.e., $u_{20} = 0$, the dynamics of tracking errors $e_{x_{1i}} = x_{1i} - x_{10}(t - \tau_{\text{coni}})$ and $e_{x_{2i}} = x_{2i} - x_{20}(t - \tau_{\text{coni}})$ becomes

$$\begin{cases} \dot{e}_{x_{1i}} = e_{x_{2i}} \\ \dot{e}_{x_{2i}} = u_{2i}(t - \tau_{\text{coni}}) \end{cases}. \quad (50)$$

The transformation (29) becomes

$$\begin{cases} e_{y_{1i}} = e_{x_{1i}} + \int_{-\tau_{\text{coni}}}^0 (-\tau_{\text{coni}} - s) u_{2i}(t+s) ds \\ e_{y_{2i}} = e_{x_{2i}} + \int_{-\tau_{\text{coni}}}^0 u_{2i}(t+s) ds \end{cases}. \quad (51)$$

After transformation, (50) becomes

$$\begin{cases} \dot{e}_{y_{1i}} = e_{y_{2i}} - \tau_{\text{coni}} u_{2i}(t) \\ \dot{e}_{y_{2i}} = u_{2i}(t) \end{cases}. \quad (52)$$

Define new variables

$$\begin{cases} e_{z_{1i}} = e_{y_{1i}} + \tau_{\text{coni}} e_{y_{2i}} \\ e_{z_{2i}} = e_{y_{2i}} \end{cases} \quad (53)$$

and we have

$$\begin{cases} \dot{e}_{z_{1i}} = e_{z_{2i}} \\ \dot{e}_{z_{2i}} = u_{2i}(t) \end{cases}. \quad (54)$$

Terminal sliding surface can be constructed as

$$s_i = e_{z_{2i}} + \alpha_1 \mu_r (e_{z_{1i}}) + \beta_1 \text{sig}(e_{z_{1i}})^{m_1/n_1} \quad (55)$$

where

$$\mu_r(e_{z_{1i}}) = \begin{cases} \text{sig}(e_{z_{1i}})^{p_1/q_1}, & \text{if } |e_{z_{1i}}| > r \\ -\beta_1/\alpha_1 \text{sig}(e_{z_{1i}})^{m_1/n_1} + k_1 e_{z_{1i}} + k_2 \text{sign}(e_{z_{1i}}) & \text{if } |e_{z_{1i}}| \leq r \end{cases} \quad (56)$$

where k_1 and k_2 are positive real numbers, $0 < r < 1$ and $k_1 \alpha_1 r + k_2 \alpha_1 = \alpha_1 r^{p_1/q_1} + \beta_1 r^{m_1/n_1}$, $\text{sig}(\cdot)^\alpha = |\cdot|^\alpha \text{sign}(\cdot)$.

To achieve fixed-time leader-following lag consensus, the controller can be designed as

$$u_{2i} = \begin{cases} -\alpha_1 p_1/q_1 |e_{z_{1i}}|^{p_1/q_1-1} e_{z_{2i}} - \beta_1 m_1/n_1 |e_{z_{1i}}|^{m_1/n_1-1} e_{z_{2i}} \\ -(\alpha_1 |s_i|^{m_2/n_2} + \beta_1 |s_i|^{m_3/n_3})^{n_1/m_1} \text{sign}(s_i), & \text{if } |e_{z_{1i}}| > r \\ -\alpha_1 k_1 e_{z_{2i}} - (\alpha_1 |s_i|^{m_2/n_2} + \beta_1 |s_i|^{m_3/n_3})^{n_1/m_1} \text{sign}(s_i) & \text{if } |e_{z_{1i}}| \leq r. \end{cases} \quad (57)$$

Theorem 4: Consider a multiagent system consisting of one leader whose dynamics is described by (3) with $u_{20} = 0$ and a group of followers whose dynamics are described by (2). Suppose that Assumptions 2 and 3 hold. Then, the distributed control scheme consisting of distributed observer (19) and terminal sliding mode protocol (57) can achieve fixed-time leader-following lag consensus for the multiagent system within finite time upper bounded by

$$T < 2 \left(\frac{1}{\alpha N^{\frac{1-m}{2}} \lambda_{\min}(H)} \frac{n}{m-n} + \frac{1}{\beta \lambda_{\min}(H)} \frac{q}{q-p} \right) + \frac{1}{\alpha_1^{n_1/m_1} (1-m_2 n_1/n_2 m_1)} + \frac{1}{\beta_1^{n_1/m_1} (m_3 n_1/n_3 m_1 - 1)} + \frac{1-r^{1-p_1/q_1}}{\alpha_1} \frac{q_1}{q_1-p_1} + \frac{1}{\beta_1} \frac{n_1}{m_1-n_1} + \frac{1}{\alpha_1 k_1} \ln \left(1 + \frac{k_1 r}{k_2} \right) + \max(\tau_{\text{coni}}). \quad (58)$$

Proof: Similar to the proof process of Theorems 1 and 2, we can prove Theorem 4 holds and the proof is omitted here.

Remark 4: From (49) and (58), it can be seen that the upper bound of the convergence time only depends on observer parameters, controller parameters, network parameters, and delay time. We can tune the observer and controller parameters or change the communication network topology to satisfy the requirement of convergence time.

Remark 5: From Theorems 3 and 4, it can be seen that whether the leader's control input is zero or not does not influence the settling time. It only affects the controller design process. Obviously, the controller design process with $u_{20} \neq 0$ is more complicated than that with $u_{20} = 0$.

IV. FURTHER DISCUSSION

A. Comparison With the Existing Literatures

In this section, we will make a comparison with the existing literatures and demonstrate the novelty of the results presented in this paper.

Most of the existing consensus protocols can achieve complete consensus. However, in some cases, complete leader-follower consensus is harmful and may result in congestion or collision, especially in some capacity-limited networks. Lag consensus can avoid these problems. The existing results on lag consensus, for example, see [42] and [43], assume identical time delay and can only achieve asymptotical lag consensus. In this paper, nonidentical time delay is considered and the proposed control protocol can achieve fixed-time leader-following lag consensus.

Fixed-time coordinated tracking for second-order multiagent system without time delay was investigated in [30]. The main differences between the control scheme presented in this paper and the control scheme presented in [30] are that a different observer and a different sliding surface are adopted and time

delay is explicitly handled in the controller design. Furthermore, the control scheme presented in [30] utilizes non-linear sinusoidal function to tackle the singularity problem. However, the singularity term in control input decays to zero when the error states approach to zero, which may reduce the convergence rate. This paper presents a novel method to tackle singularity problem, which guarantees fast convergence rate. Finally, the control scheme presented in [30] cannot eliminate chattering and convergence error simultaneously. The control protocol presented in this paper is continuous, thereby eliminating chattering phenomenon. When the error states are outside the region $|e_{z_{1i}}| \leq r$, they will enter into the region within fixed-time and after the system enter into the region, they will achieve exact convergence to the origin within finite time. Therefore, the proposed control protocol can eliminate chattering and convergence error simultaneously.

Most of the existing fixed-time control schemes are designed for first-order integrator system and direct extension to second-order case has encountered a challenging problem, i.e., singularity problem [26]. Recently, some progress has been made to address this problem and achieve fixed-time stabilization for second-order system (32). In [26] and [53], a fixed-time terminal sliding mode controller was designed using a non-linear sinusoidal function to overcome the singularity problem. In [23], a fast fixed-time terminal sliding mode controller was designed using saturation function to avoid singularity problem. It is worth noting that these methods limit the control input but prolong the convergence time and cannot give accurate estimation for the traveling time through the area where the functions that are used to overcome singularity take effect, which results in inaccurate estimation for total convergence time and the difficulties in selecting controller parameters to ensure desired convergence time. In addition, the controllers they designed are too complicated. In this paper, a novel control protocol is proposed. In (35) and (57), if $|e_{z_{1i}}| > r$, the control protocol includes singularity term but its amplitude is bounded. After the system enters into $|e_{z_{1i}}| < r$, there is no singularity term in the control protocol. Therefore, the proposed control protocol overcomes singularity problem. Moreover, the proposed control protocol is simple, can give exact estimation for the convergence time bounds and ensures bounded control input without sacrificing convergence time.

The consensus problem of multiagent system is investigated in many literatures. For example, in [54], some consensus criteria were derived using a general Halanay-type inequality with time-varying coefficients and an improved comparison principle. In [55], partial mixed impulses were utilized to achieve leader-following consensus. However, the methods presented in these papers can only achieve exponential consensus, which implies exact consensus cannot be achieved within finite time. In this paper, we design a fixed-time distributed observer and a novel nonsingular fixed-time terminal sliding mode protocol, and use an extension of the Artstein's reducing transformation to achieve fixed-time consensus. The presented method can achieve exact consensus within finite time and the convergence time is upper bounded by a constant independent of initial condition.

B. About Time Delay

In this section, we will discuss some issues about time delay, including the reasonability of the assumption, the effect of time delay on controller design and the delay size that the controlled system can tolerant.

In this paper, we consider known constant input delay. However, it can also be extended to unknown constant and time-varying delay. For unknown constant input delay, the estimation technique presented in [56] can be used to estimate the delay. For time-varying delay, as shown in [57], we can consider the worst case when selecting the input delay.

Time delay influences the design of controllers in the following aspects. First, for delayed system, we use the state variables $e_{z_{1i}}$ and $e_{z_{2i}}$ after Artstein's transformation to design controller, whereas for delay-free system, we can use the original state variables $e_{x_{1i}}$ and $e_{x_{2i}}$ to design controller. Second, the delay time is added into the convergence time, which means that larger control gains are needed to achieve consensus within the same settling time as the delay-free one. Third, exact leader-following lag consensus can be achieved for delayed system while exact leader-following consensus can be achieved for delay-free system.

Theoretically, the proposed control protocol can achieve fixed-time leader-following lag consensus regardless of delay size. In many practical systems, the system states are required to operate within a certain range to ensure safe and stable operation. Otherwise, it will cause damages to the system and even result in system instability. The values within this range are called acceptable values and any values outside this range are called unacceptable values. Therefore, practically, the time delay is required to be bounded such that the system states cannot escape to an unacceptable value before the controller can affect the system.

V. SIMULATION RESULTS

In our simulation, we consider coordinate lag tracking problem for a group of single-link robotic manipulators [7]. The leader node generates prescribed trajectory and its dynamics can be described by

$$\ddot{q}_0 = u_{20} \quad (59)$$

where $\ddot{q}_0$ and u_{20} are the angular acceleration of the prescribed trajectory and control input, respectively. Denote $x_{10} = q_0$ and $x_{20} = \dot{q}_0$, and the leader's dynamics becomes (3).

The followers are three single-link robot arms with the dynamics described by

$$J_i \ddot{q}_i + B_i \dot{q}_i + M_i g l_i \sin(q_i) = u_i \quad (60)$$

where q_i and $\dot{q}_i$ are angle and angular velocity of the i th link, J_i is the total rotational inertia of the link and the motor, B_i denotes damping coefficient, M_i represents the total mass of the link, g is the gravitational acceleration, and l_i is the distance from the joint axis to the link center of the mass for node i . In order to achieve fixed-time lag consensus, the control input of

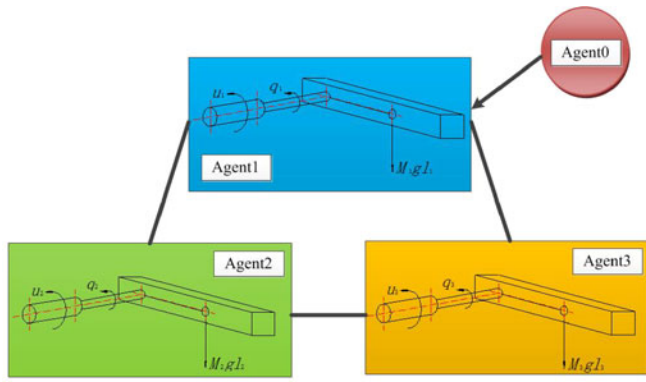


Fig. 1. Interaction graph.

the followers is designed as

$$u_i = u_l^i + u_g^i = B_i \dot{q}_i + M_i g l_i \sin(q_i) + J_i u_{2i}(t - \tau_{\text{coni}}) \quad (61)$$

where the control input is decomposed into the local control term $u_l^i = B_i \dot{q}_i + M_i g l_i \sin(q_i)$ and the global control term $u_g^i = J_i u_{2i}(t - \tau_{\text{coni}})$, respectively. The local control term is designed based on the agent's own output vector and the global control term utilizes information from its neighboring agents and has time delay τ_{coni} .

With the control input (61), the followers' dynamics become

$$\ddot{q}_i = u_{2i}(t - \tau_{\text{coni}}). \quad (62)$$

We denote $x_{1i} = q_i$ and $x_{2i} = \dot{q}_i$, and the followers' dynamics become (2). The systems are linked through a connected undirected interaction graph as shown in Fig. 1.

Next, two simulation examples are provided to demonstrate the fixed-time lag consensus performance of the presented protocols. In the first example, the control input of the leader agent is selected as $u_{20}(t) = -0.1 \sin(t)$ with the initial condition $(x_{10}, x_{20}) = (0, 0.1)$. The initial conditions for the follower agents are chosen as $(x_{11}(0), x_{21}(0), x_{12}(0), x_{22}(0), x_{13}(0), x_{23}(0)) = (0.4, 0.25, 0.2, 0.2, 0.5, 0.3)$. The observer parameters are selected as $\alpha = \beta = 10, m = 5, n = 3, p = 3, q = 5, \lambda = 0.1$ and the controller parameters are selected as $p_1 = 3, q_1 = 5, m_1 = 5, n_1 = 3, m_2 = 7, n_2 = 5, m_3 = 7, n_3 = 3, \alpha_1 = 0.5, \beta_1 = 0.5, r = 0.396, k_1 = 1.71, k_2 = 0.11$. Input delay are selected as $\tau_{\text{con1}} = 0, \tau_{\text{con2}} = 0.5, \tau_{\text{con3}} = 1$. We employ (50) to calculate the upper bound of convergence time and the computation result is $T < 24.5938$. We implement the simulations with the protocol (35) and plot the time responses of error states and system states in Fig. 2. It can be seen from Fig. 2 that the agent 1 achieves exact leader-following consensus, whereas agent 2 and agent 3 achieves exact leader-following lag consensus. Now, we run simulations under different initial conditions and obtain the actual convergence time. The results are shown in Fig. 3. As can be seen from Fig. 3, the convergence time of the proposed protocol is bounded by a constant as the logarithm of the norm of initial condition grows. The results verify the proposed control protocol achieves fixed-time consensus.

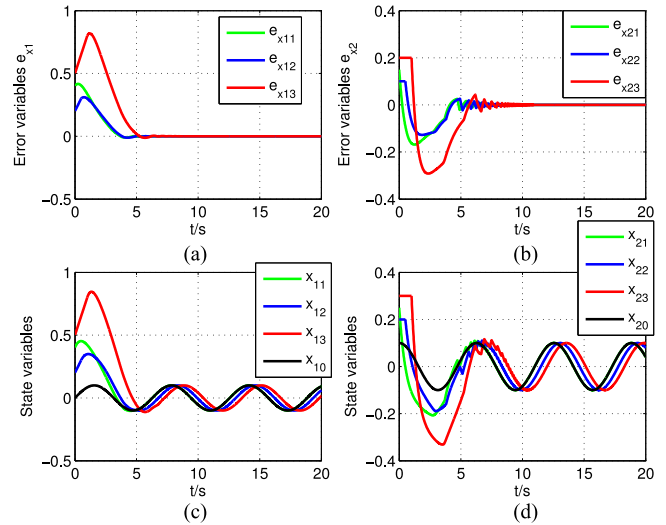
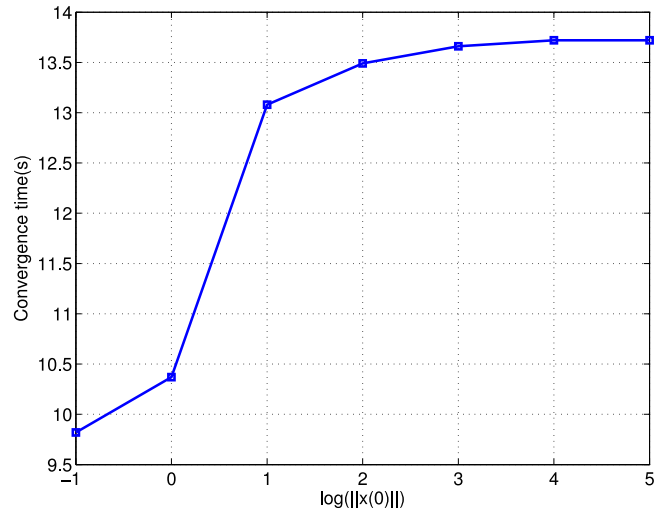
Fig. 2. Time response of error states and system states: (a) $e_{x_{1i}}$; (b) $e_{x_{2i}}$; (c) x_{1i} ; and (d) x_{2i} .

Fig. 3. Convergence time versus the logarithm of norm of initial condition.

In the second example, the leader's control input is assumed to be zero and the initial condition is selected as $(x_{10}, x_{20}) = (0, -0.2)$. The controller parameters, observer parameters and the initial conditions for the followers are selected to be the same as the first example. From (59), the upper bound of convergence time can be estimated as $T < 24.5938$. We implement the simulations with the protocol (58) and the results are shown in Figs. 4 and 5. As shown in Fig. 4, exact leader-following consensus for agent 1 and exact leader-following lag consensus for agents 2 and 3 are achieved within the same settling time as example 1. It can be seen from Fig. 5 that the convergence time of the proposed protocol is bounded by a constant as the logarithm of the norm of initial condition grows, which validates the proposed control protocol achieves fixed-time consensus. Comparing Figs. 4 and 5 with Figs. 2 and 3, we may find the convergence time without the leader's control input is the same

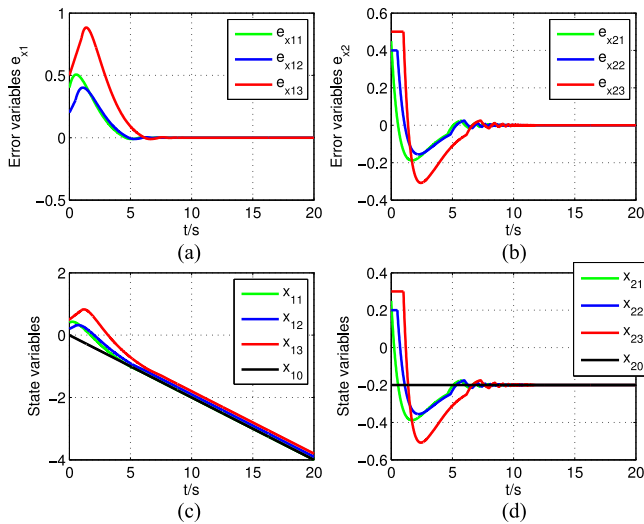


Fig. 4. Time response of error states and system states: (a) e_{x1i} ; (b) e_{x2i} ; (c) x_{1i} ; and (d) x_{2i} .

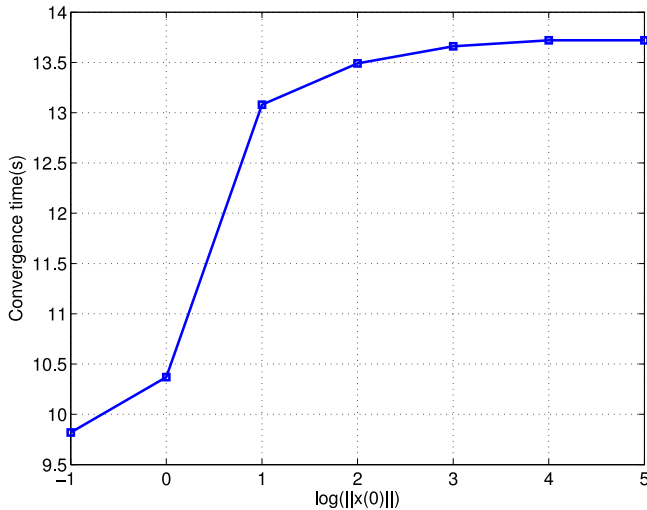


Fig. 5. Convergence time versus the logarithm of norm of initial condition.

as that with the leader's control input, which is confirmed by Theorems 3 and 4 and clarified in remark 5.

VI. CONCLUSION

This paper investigated fixed-time leader-following lag consensus problem for second-order multiagent systems with input delay. Fixed-time distributed observer was first constructed to obtain the leader's states. By utilizing an extension of Artstein's reducing transformation, the delayed error system was transformed into a second-order system without time delay and then a newly designed nonsingular terminal sliding mode protocol was presented to achieve fixed-time consensus. The presented sliding mode controller can avoid singularity, eliminate chattering, and achieve exact convergence. It was proved that the presented protocol can achieve fixed-time leader-following lag consensus.

In particular, the convergence time was upper bounded by a constant that depends only on observer parameters, controller parameters, network parameters, and delay time. This makes it possible to determine the upper bound of convergence time offline regardless of initial condition. Application into coordinated lag tracking control of single-link robotic manipulators was also provided and simulation results demonstrated the effectiveness of the proposed protocol.

Recently, stochastic delayed multiagent systems have attracted great attention. Some interesting results about finite-time synchronization of stochastic delayed complex network have been obtained, such as [58]. Based on [54], [55], and [58], in our future work, we will extend the results of this paper to fixed-time leader-following consensus of stochastic delayed multiagent systems.

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